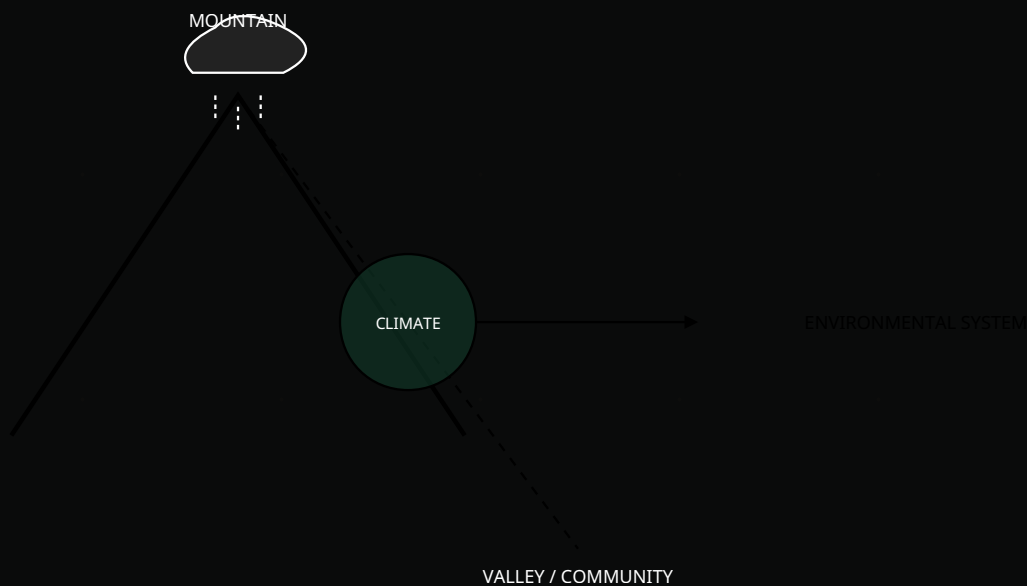


CLIMATE & ENVIRONMENT

“Where mountains, water, weather and life meet.”

The mountains shape the landscape. The valleys collect water. And climate moves through both. Temperature, rainfall, snowfall, seasonal variation, vegetation and water availability are not separate stories. Together, they form the environmental system in which life in Orakzai takes place.

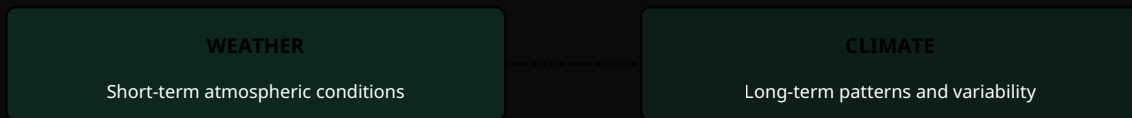
Understanding the environmental context of Orakzai requires recognizing it as an ecologically transitional zone between subtropical and temperate regions. The interaction between rugged topography and atmospheric patterns determines the rhythm of agricultural life, the distribution of flora, and the challenges faced by local communities.¹²



“Conceptual model of the environmental system: Mountain » Climate » Valley » Life.”

CLIMATE VS. WEATHER

It is essential to distinguish between weather and climate. While weather describes short-term atmospheric conditions—such as a specific rainfall event or a cold morning—climate refers to the long-term patterns and variability observed over decades. In Orakzai, the climate is the predictable yet varying framework within which daily weather occurs.²³



“The fundamental distinction between short-term weather and long-term climate.”

ORAKZAI'S MOUNTAIN CLIMATE

Orakzai's mountainous terrain creates significant elevation differences that directly influence local environmental conditions. Generally, higher elevations in the west experience cooler temperatures and greater potential for snowfall, while the lower valleys in the east remain relatively milder. This physical relationship shapes the ecological transition from subtropical to temperate zones.¹⁴



“Physical relationship between elevation and climatic conditions.”

SEASONAL CYCLE

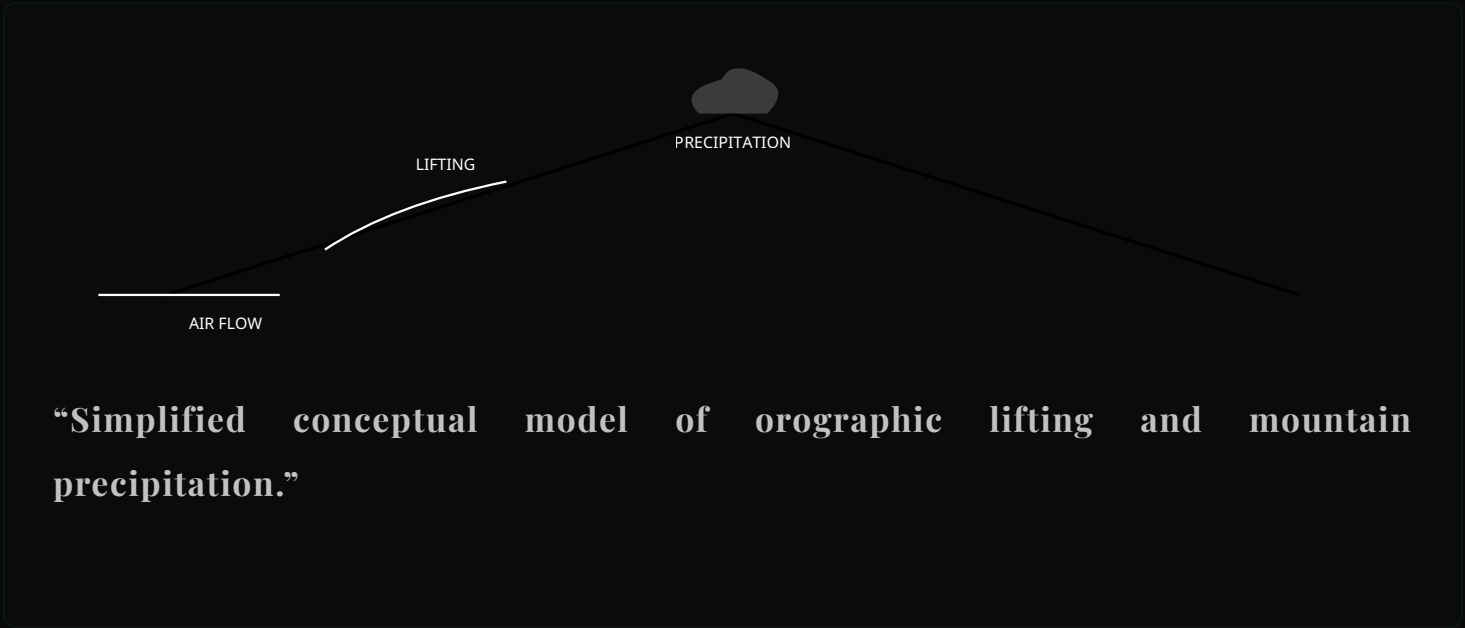
The regional seasonal pattern is characterized by four distinct periods. Winter brings cold temperatures and significant snowfall to higher reaches like the Tirah Valley, while Summer is temperate in the upper basins but influenced by the seasonal monsoon. These transitions dictate the timing of agricultural activity and water availability.²⁴



“The circular rhythm of the four seasons in Orakzai.”

PRECIPITATION: RAINFALL & SNOWFALL

Rainfall and snowfall in Orakzai are driven by complex mechanisms. Winter precipitation is primarily associated with **Western Disturbances**, while summer rains are influenced by the **Monsoon**. Mountains play a critical role through **Orographic Lifting**, where rising air cools and condenses, leading to localized precipitation.²³



CLIMATE CHANGE & ENVIRONMENTAL RISK

Regional and national climate evidence documents significant trends, including rising temperatures and changing rainfall variability. For Orakzai, these changes manifest as increased unpredictability of seasonal rains and heightened risks of rapid runoff leading to valley floods. Adaptation to these future risks is a critical component of regional resilience.³⁵

FLOOD RISK

Intense rainfall on steep terrain produces rapid runoff, increasing the threat of sudden floods in valley floors.⁵

DROUGHT RISK

Variability in precipitation can lead to periods of water stress, impacting agriculture and local livelihoods.³

WARMING TRENDS

Regional warming affects snowmelt timing and the ecological balance of the transitional zones.³

SOIL EROSION

Heavy rainfall events on mountainous slopes contribute to soil loss and environmental degradation.¹

The environmental richness of Orakzai is evidenced by its diverse flora, with 123 identified wild plant species. These include traditionally significant medicinal plants like **Daphne papyracea** and **Olea europaea**. However, 37% of these species are classified as highly vulnerable, highlighting the need for conservation and the preservation of tribal ecological knowledge.¹

123 WILD PLANT SPECIES

37% HIGHLY VULNERABLE

“The scale of biodiversity and the urgent challenge of conservation.”

AUTHOR'S REFLECTION

Climate is the breath of the mountains. It is the snow that blankets the Tirah and the monsoon that feeds our orchards. We have lived in harmony with these cycles for generations, but the world is changing. The rains are less predictable, and the heat is rising. To protect Orakzai is to protect its environment—not as a wilderness to be locked away, but as a living system that sustains our people and our identity.

“The environment is the silent partner in every Orakzai story.”

AUTHOR'S SYNTHESIS

Note: District-level long-term climate observations are limited in the publicly accessible sources reviewed. Regional and national trends are used as indicators where district-specific measurements are unavailable.